



PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—PRELIM PAPER 2—

27 September 2023

Examiner: Mr. D. Grace

Internal Moderator: Mr. A Meintjes

External Moderator: Mr. D. Harvey

TIME: 3 hours

MARKS: 150 Marks

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. Write your name and class in the space provided on the answer book.
 2. Please check that your question paper is complete.
 3. Read the questions carefully and answer ALL the questions in your answer booklet.
 4. It is in your own interest to write legibly and to present your work neatly.
 5. You are provided with a 1 : 50 000 topographic map 2930CA MERRIVALE and an orthophoto 2930CA 5 MERRIVALE of a part of the mapped area.
 6. You must hand the topographic map and orthophoto map to the invigilator at the end of this examination session. You may only use pencil on these maps. Please erase any pencil markings on the map before handing it back in.
 7. Show ALL calculations and formulae, where applicable. Marks will be allocated for these.
 8. Indicate the unit of measurement or compass direction in the final answer of calculations, e.g. 10 km; 2.1cm; west of true north.
 9. You may use a non-programmable calculator.
 10. You may make use of a magnifying glass.
 11. The area demarcated in RED/BLACK on the topographic map represents the area covered by the orthophoto map.
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SECTION A – Rural and Urban Settlement

QUESTION 1: Short Questions

- 1.1** Match the term/concepts below to the statements that follow. Write **ONLY** the term/concept next to the question number. Example 1.1.8 Commercial zone.

urban hierarchy; threshold population; urban profile; invasion;
break-of-bulk point; urban sprawl; facadism; site; gentrification; ghetto.

- | | | |
|-------|--|----------|
| 1.1.1 | Minimum number of people needed to make a business profitable. | (1x1)(1) |
| 1.1.2 | Replacement of the original type of land use with another. | (1x1)(1) |
| 1.1.3 | Shows the height of the buildings of a city. | (1x1)(1) |
| 1.1.4 | The shapeless uncontrolled expansion of the city. | (1x1)(1) |
| 1.1.5 | When the old, original front or façade of a building is retained and a new development is built behind the old façade. | (1x1)(1) |
| 1.1.6 | Ranking of urban settlements according to their function. | (1x1)(1) |
| 1.1.7 | The actual place where a settlement grew. | (1x1)(1) |

[7]

- 1.2** Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

- | | | |
|-------|--|----------|
| 1.2.1 | The largest city type with significant economic power is known as a... | (1x1)(1) |
| | A megalopolis.
B smart city.
C conurbation.
D metropolis. | |
| 1.2.2 | The smallest rural settlement is a... | (1x1)(1) |
| | A village.
B hamlet.
C town.
D isolated farmstead. | |

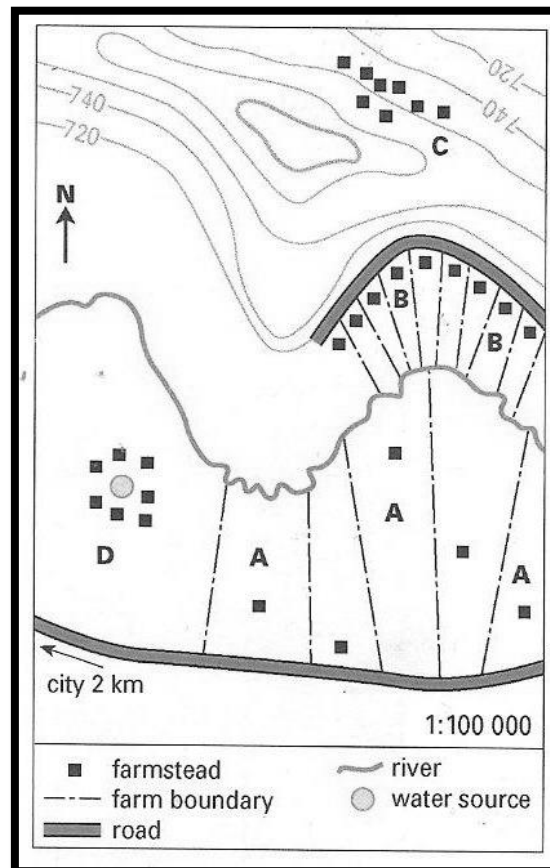
- 1.2.3 The largest rural settlement, is a... (1x1)(1)
- A town.
 - B hamlet.
 - C village.
 - D isolated farmstead.
- 1.2.4 The smallest urban settlement is a... (1x1)(1)
- A city.
 - B town.
 - C village.
 - D metropolis.
- 1.2.5 A very small grouping of farmsteads without any services or shops is a... (1x1)(1)
- A minor country town.
 - B village.
 - C hamlet
 - D central place.
- 1.2.6 An example of specialised city is... (1x1)(1)
- A Johannesburg.
 - B the Vatican.
 - C De Aar.
 - D Durban.
- 1.2.7 An example of a gateway/gap town is... (1x1)(1)
- A Worcester.
 - B Pretoria.
 - C Phalaborwa.
 - D Saldanha Bay.
- 1.2.8 The transitional zone is known for... (1x1)(1)
- A having mixed land use.
 - B undergoing change.
 - C being on the edge of the commercial zone.
 - D all of the above.

(8x1)(8)

[8]**[15]**

QUESTION 2: Classification of Rural Settlements

- 2.1 Refer to the diagram below and answer the questions that follow. Match each of the rural settlements labelled A, B, C and D with one of the following statements:



- 2.1.1 Irregular and nucleated pattern. (1x1)(1)
- 2.1.2 Irregular and dispersed pattern. (1x1)(1)
- 2.1.3 Nucleated pattern and linear shape. (1x1)(1)
- 2.2 **Describe** ONE factor that may have influenced the pattern and shape of each settlements labelled A-D. (4x2)(8)
- 2.3 **Differentiate** between the terms site and situation. (2x2)(4)

[15]

QUESTION 3: Case study

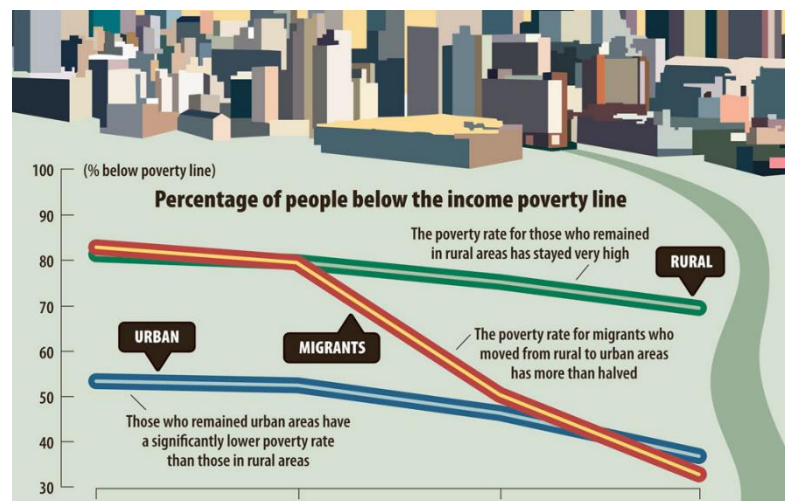
Refer to the article below and answer the questions that follow.

Rural migrants better off in cities.

Moving to a city in search of work seems to pay off for many poor rural South Africans.

Data that track changes over time indicate that as many as 385 000 people were lifted from poverty between 2008 and 2014 after moving from rural to urban areas — their poverty levels were halved, together with a fall in

unemployment. So government ambivalence about urbanisation should be replaced by a more positive and proactive approach. The fate of people who uproot themselves and migrate to cities is widely discussed. Conventional wisdom says that most end up living in shack settlements, where they struggle to make ends meet because they lack the skills to compete for the paltry jobs available. Their frustrations give rise to protests, land invasions and other forms of antisocial behaviour. Another scenario is that, by sheer determination, they manage to grasp some kind of urban livelihood and gradually claw their way up the job ladder. Historically, there has been a strong relationship between urbanisation and economic development in many parts of the world. Cities function as catalysts for prosperity; they can generate economies of scale and scope, reduced logistics costs and access to shared services, specialised suppliers, thick labour markets and knowledge spillovers. But the benefits of density and proximity are not automatic. Many African cities suffer from negative externalities such as congestion, infrastructure bottlenecks and pollution.



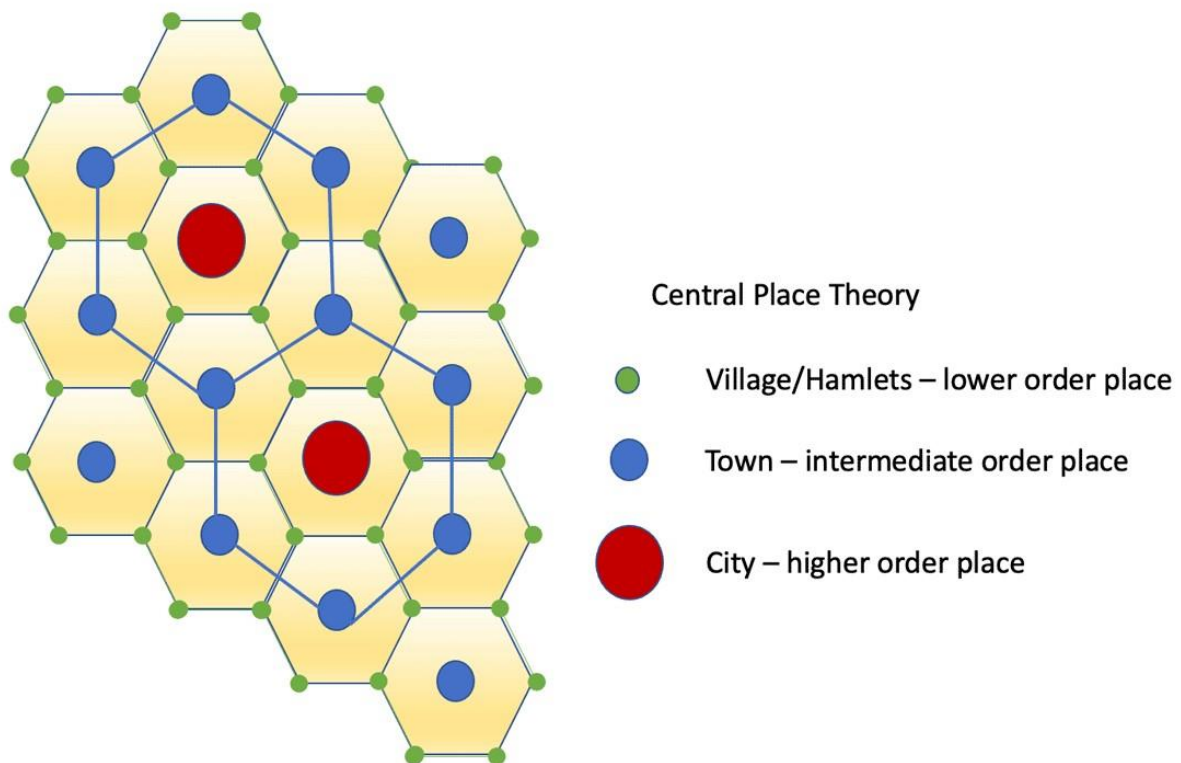
- 3.1 **What** do you understand by the sentence “so government ambivalence about urbanisation should be replaced by a more positive and proactive approach.”? (1x2)(2)
- 3.2 **Distinguish** between rural depopulation and selective outward migration. (2x2)(4)

- 3.3 With reference to the article, **discuss** the ONE positive and ONE negative consequence of rural-urban migration. (2x2)(4)
- 3.4 **Provide** and **explain** ONE cause of rural depopulation. (1x2)(2)
- 3.5 **Name** THREE consequences of rural depopulation for the rural settlement. (3x1)(3)

[15]

QUESTION 4: Central place theory

4. Refer to the diagram below and answer the questions that follow.



- 4.1 **What** is an urban hierarchy? (1x2)(2)
- 4.2 **Who** developed the idea of central place theory? (1x1)(1)
- 4.3 **Why**, according to Central Place theory, do we find more lower order places than higher order places? (1x2)(2)
- 4.4 **Name** ONE low and ONE higher order function or service suggested by the Central Place theory. (2x1)(2)

- 4.5 In a paragraph of approximately **EIGHT lines**, discuss the **three** main ideas that underpin the Central Place theory and **provide and explain ONE** South African example where this applies.

(4x2)(8)

[15]

TOTAL SECTION A: [60]

SECTION B – Economic Geography

QUESTION 5: Short Questions

- 5.1 Provide the correct geographic term or phrase that fits each definition or explanation below. Write only the number and the correct answer.

Column A	Column B
5.1.1 Centripetal force	A: The side view of a city.
5.1.2 Urban profile	B: When old houses close to the city centre are modernised and improved.
5.1.3 Range	C: Changing the existing structure of land tenure; redistribution of land.
5.1.4 GEAR strategy	D: A strategy adopted by the government in 1994 to solve social and economic problems.
5.1.5 RDP	E: Maximum distance that people travel to buy goods.
5.1.6 Gap town	F: Land owned by the community.
5.1.7 Gentrification	G: A gateway into the interior.
5.1.8 PWV complex of Gauteng	H: A plan to promote growth, employment and redistribution.
	I: A force that attracts and keeps people in an urban settlement.
	J: One of the core industrial regions in South Africa.
	K: A force that repels and pushes people out an urban settlement.

[8]

- 5.2 **Choose the correct concept/term from those given between brackets to make the statements TRUE.** Write only the concept/term next to the question number. Example: 2.2.8 Industry.
- 5.2.1 **(Footloose/Ubiquitous)** industries that can be located anywhere. (1x1)(1)
- 5.2.2. An industry that produces less pollution and handles small amounts of raw materials e.g. **(Sasol/bakery)**. (1x1)(1)
- 5.2.3 Industries located close to the consumer are known as **(market/raw material)** orientated industries. (1x1)(1)
- 5.2.4 Industries that produce a lot of pollution and are located near bulk transport networks are known as **(break-of-bulk/heavy)** industries. (1x1)(1)
- 5.2.5 Industries that remain in the same location even though the original reasons for being there are gone are referred to as **(industrial inertia/hinterland)**. (1x1)(1)
- 5.2.6 Mineral **(by-products/benefication)** add value to a mineral. (1x1)(1)
- 5.2.7 **(Multiplier effects/Link industries)** are referred to as additional economic developments that stem from the mining industry. (1x1)(1)

[7]

QUESTION 6: Agriculture

6. Refer to the image below and answer the questions that follow.

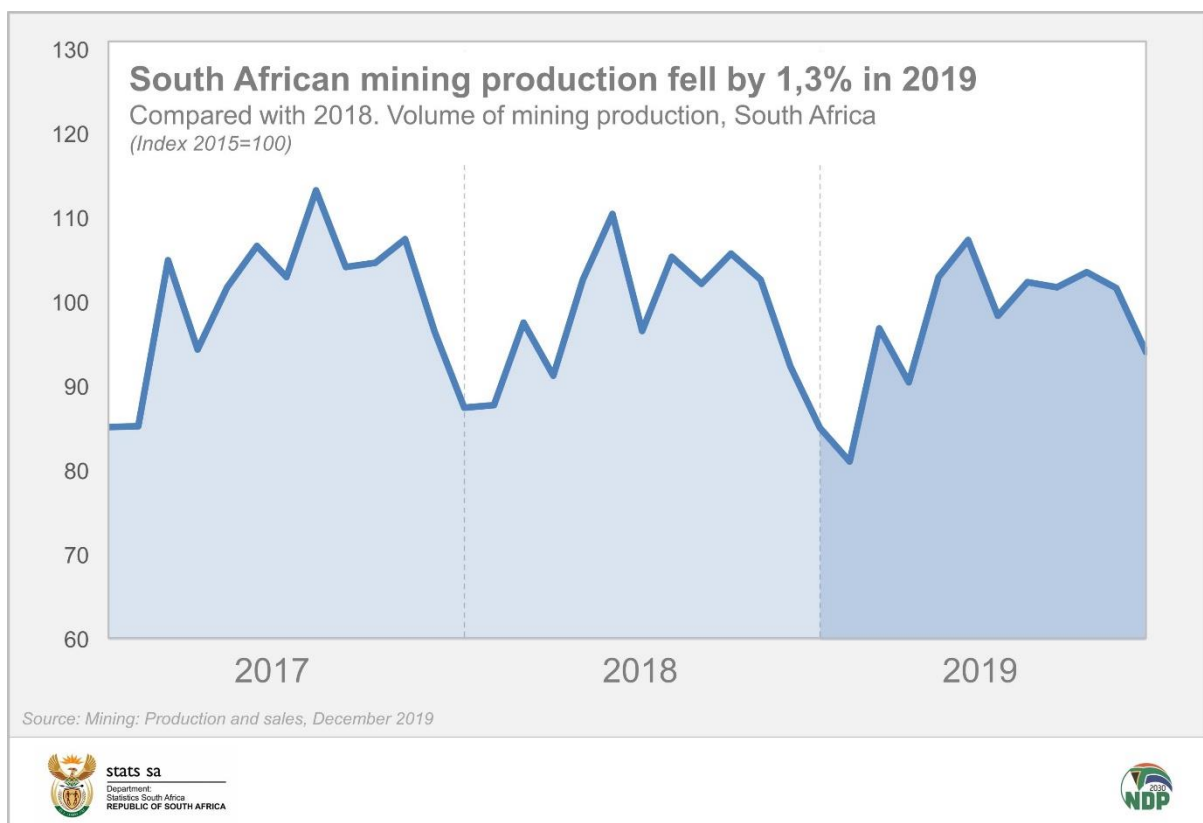


- 6.1 **Name the TWO** types of farming practised in South Africa. (2x1)(2)
- 6.2 **What** is the message in the cartoon? (1x2)(2)
- 6.3 **Provide and explain TWO** factors that favour farming in South Africa. (2x2)(4)
- 6.4 **What** is food security? (1x2)(2)
- 6.5 **What** is the link between poverty and food insecurity? (1x2)(2)
- 6.6 **List THREE** factors hindering agriculture in South Africa. (3x1)(3)

[15]

QUESTION 7: Mining

7. Refer to the figure below showing a graph and answer the questions that follow.



- 7.1 Which **TWO** years was South Africa's mining production the highest? (2x1)(2)
- 7.2 How has the discovery and extraction of minerals benefited South Africa? **Provide and explain TWO** ways. (2x2)(4)
- 7.3 List **THREE** factors that favour mining in South Africa. (3x1)(3)
- 7.4 What does the above graph suggest about the future of mining in South Africa? (1x2)(2)
- 7.5 Why should South Africa be concerned about the future of mining? **Provide and explain TWO** reasons. (2x2)(4)
- [15]

QUESTION 8: Strategies for Industrial Development

8. Refer to the cartoon below and answer the questions that follow.



- 8.1 In the year 1996 the South African government started an ambitious housing development plan. List **TWO** reasons that it was based on. (2x1)(2)
- 8.2 What does BBB-EE stand for? (1x1)(1)

8.3 **Explain** the humour in the cartoon. (2x2)(4)

8.4 In the context of South Africa's post-Apartheid transformation, **elaborate** on the fundamental principles and goals of the GEAR policy, focusing on TWO crucial elements. Following this, **assess** the effectiveness of the BEE policy in promoting black economic empowerment.

(4x2)(8)

[15]

TOTAL SECTION B: [60]

SECTION C - Mapwork

RESOURCE MATERIAL

1. 2930CA MERRIVALE and an orthophoto 2930CA 5 MERRIVALE

GENERAL INFORMATION ON MERRIVALE



Merrivale rests not far from the banks of the Howick Dam.

The Main road and the Umgeni River act as dividers. But tacked on to the southern tip of Howick as it is, Merrivale might have its own train station and accompanying

old-style character railway houses, but today it merges seamlessly with the larger Howick.

Just alongside Merrivale is the [Sakabula Golf and Country Estate](#), a major draw card to the area for golfers – the only par 73 rated golf course in the province with two very different nine hole courses, set in a rural environment.

There are also a series of falls in the area. [Howick Falls](#), after which the town is named, rush 95m down dolerite cliffs on the Umgeni river, whilst Cascade Falls, Shelter Falls and Karkloof Falls all lie within the vicinity of Howick, and make a good day trip.

QUESTION 9: Calculations

9.1 Provide the grid reference of the map directly South of **2930CA**. (1x2)(2)

9.2 **Calculate** the straight line distance between 6 and 7 on the orthophoto in km. (1x2)(2)

Use the formula:

$$\frac{\text{Distance on a map} \times \text{scale}}{100\ 000}$$

9.3 **Draw** a simple cross section between spot height 1422 and spot height 1411 in block E4. (1x2)(2)

9.4 **Calculate** the magnetic bearing from spot height 1411 to spot height 1422 in block E4.

Use the formula:

$$\text{True Bearing} + \text{Magnetic Declination} = \text{Magnetic Bearing} \quad (4x1)(4)$$

[10]

QUESTION 10: Interpretation

10.1 In **which** province is Merrivale located? (1x1)(1)

10.2 **What** type of economic activity is predominate in B5? **Provide** a reason for your answer. (2x1)(2)

- 10.3 **What** is the dominant street pattern in block C1 of the Topographic map? **Name** ONE benefit of this pattern. (2x1)(2)
- 10.4 **Account** for the location of the sewerage works in block C2. (2x1)(2)
- 10.5 **Which** area is wealthier between Cedarge (blocks E1 & E2) and Mpophomeni (blocks D2 and D3)? **Provide** a reason for your answer. (2x1)(1)
- 10.6 Urbanisation has been increasing in the Merrivale region lately, particularly around block C1. **Provide** ONE piece of evidence that supports this statement and ONE suggestion to further increase urbanisation in the future. (2x2)(4)

[12]

QUESTION 11: GIS

- 11.1 Consider the Orthophoto map.
- 11.1.1 **Is** feature 6 on the orthophoto map and example of a point, line or polygon feature? (1x1)(1)
- 11.1.2 **Which** format, raster or vector, would be best for showing categories of data such as elevation? (1x1)(1)
- 11.1.3 **Name** the GIS concept that assigns a value or characteristic to each geographical feature. (1x1)(1)
- 11.1.4 **Name** ONE layer that would have likely been considered by GIS consultants before developing the powerlines in this area. (1x1)(1)
- 11.1.5 **Explain** how you could use GIS in managing the buffer zone around the dams in block C1 and C2 of this map. (2x2)(4)

[8]

TOTAL SECTION C: [30]

GRAND TOTAL:

150 MARKS